

Implementation Roadmap – StockWise Project

Overview

The Implementation Roadmap outlines the development phases of the StockWise inventory management system. It defines the tasks, milestones, and timeline for completing the project efficiently and systematically.

Phase 1 – Project Initiation (Weeks 1–2)

- Define project objectives and scope
- Assign team roles and responsibilities
- Set up Jira board and repository structure
- Prepare initial project documents (project plan, overview, requirements template)

Phase 2 – Requirements Analysis (Weeks 3–4)

- Collect functional and non-functional requirements
- Create UML diagrams for system structure and workflows
- Identify potential risks and mitigation strategies
- Confirm system architecture and technology stack

Phase 3 – System Design (Weeks 5–6)

- Design database schema (PostgreSQL)
- Prepare API specifications (Spring Boot backend)
- Design frontend UI/UX (React.js)
- Finalize security design (RBAC, authentication, encryption)

Phase 4 – Development (Weeks 7–10)

- Implement backend APIs (CRUD for Products, Categories, Alerts, Users)
- Integrate frontend dashboard with backend APIs
- Implement stock monitoring and alert system
- Conduct unit testing during development

Phase 5 – Testing and Validation (Weeks 11–12)

- Perform integration testing between frontend and backend
- Execute manual testing scenarios
- Identify and fix bugs or performance issues
- Prepare final test reports

Phase 6 – Final Delivery (Weeks 13–14)

- Deploy backend and frontend to production
- Prepare final documentation and technical reports
- Upload all project deliverables to GitHub repository
- Present project using GitHub as the showcase